

MINUTES OF MEETING (MOM)

Team 8

3D Printing Cost Estimation Platform

Meeting Details

Project: 3D Printing Cost Estimation Platform
Week: Week 8
Date: Friday, March 27, 2026
Time: 4:00 PM IST
Location: CIE, Ground Floor
Client: Anil Kumar Upadhyay, Five Fingers Innovative Solutions
Client Email: aku1974india@gmail.com
Client Phone: 9949497560

Team Members Present

- Abhiroop Pullepu
- Mehrish Khan
- Ronith Menneni
- Sai Hasini
- Ruschita Guddeti

Notes

- Follow-up to the March 16 meeting where OTP verification, admin configuration, and CuraEngine parameter passthrough were discussed and planned.
- Focus: review implementation progress, small bugs discovered during testing, and adjustments requested by the client.

1 Progress Since March 16

1.1 Authentication and OTP Flow

- Backend `/api/auth/verify-otp` endpoint implemented with 6-digit OTP, 10-minute expiry, and single-use tokens.
- OTP emails sent via Mailtrap using the client-provided token.
- Frontend OTP entry screen integrated into login flow; errors shown for invalid/expired OTP.

1.2 Admin Configuration Panel

- Admin UI for editing:
 - Material cost per gram per material type (PLA, PETG, etc.).
 - Machine time rate (INR per minute).
 - Waste and failure factors (no longer hard-coded).
- Changes persisted to the `adminconfig` table and reflected in subsequent estimates.

1.3 Slicing Parameters & Failure Logs

- User-selectable parameters (infill, layer height, supports, model fill type) being passed through to the slicing service.
- Failure logging now includes the full parameter set, job ID, and timestamp, visible in the admin view.

2 Bugs and Issues Identified

2.1 OTP-Related Issues

- **Issue: OTP mismatch after resend**
 - Symptom: In a few cases, user entered a valid OTP but received “invalid OTP” after quickly requesting a resend.
 - Root cause: Old OTP not invalidated before generating a new one.
 - Fix: Ensure that generating a new OTP marks all previous ones for that user/session as expired.
 - Status: Fix implemented and retested.
- **Minor UX Issue: Confusing timer**
 - Symptom: Timer label showed “Resend in 0s” but button still disabled briefly.
 - Fix: Align UI timer with backend cooldown; disable button until timer strictly > 0 is false.
 - Status: Fixed.

2.2 Admin Panel Issues

- **Issue: Unsaved changes warning missing**
 - Symptom: Admin could navigate away from the page after editing values without any warning.
 - Fix: Added unsaved-changes confirmation dialog when form is dirty.
 - Status: Fixed.
- **Issue: Extreme values accepted**

- Symptom: Admin could enter unrealistic values (e.g., waste factor of 10.0).
- Fix: Added validation ranges (e.g., waste/failure factors limited to reasonable bounds).
- Status: Fixed; ranges documented for the client.

2.3 Printability Analysis Tuning

- Thin-wall detection still returns a few false positives on complex models, but significantly fewer than before.
- Team agreed to log borderline cases and refine heuristics in a subsequent sprint rather than blocking the current one.

3 Client Feedback

- Client is satisfied that OTP adds a second layer of protection against credential sharing.
- Admin configuration form meets initial expectations; client requested a short inline help text explaining each field.
- Asked the team to ensure that failure logs can be exported (CSV) in a later iteration.

4 Next Steps

Team Deliverables (Due: April 3, 2026)

Owner	Task
Abhiroop Pullepu	Harden OTP flow (rate limiting, lockout after multiple failed attempts); document security assumptions.
Mehrish Khan	Connect admin-configured waste/failure factors to cost engine tests; update unit tests for new ranges.
Ronith Menneni	Add tooltip/help text and form validation messages to admin panel; ensure mobile responsiveness.
Sai Hasini	Polish OTP UI (error states, focus handling) and test across browsers.
Ruschita Guddeti	Draft a one-page “Release 1.1” feature summary for client sign-off (OTP, admin panel, improved printability).

Client Deliverables (Due: April 1, 2026)

- Confirm final default values for machine time rate and waste/failure factors to be used when admin config is reset.

Minutes prepared by: Team 8
Date: March 27, 2026